



Technical Service Bulletin

Technical Service Bulletin: TSB180052

Released Date: 21-May-2018

Changes to Base Engine Timing

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

- ISX15 CM2250
- ISX15 CM2350
- QSX15 CM2250
- QSX15 CM2350
- X15 CM2350

Description of Change

This document announces the release of revised procedures for Base Engine timing (001-088) and Gear Train Backlash (001-055).

Major changes include:

- A new nominal target of 0.004" for both adjustable fuel pump idler gear meshes
- Used bolts are to be discarded for fuel pump idler gear when removing Gear
- New bolts with pre-applied thread-locker are to be used for fuel pump idler gear when reinstalling gear
- Fuel pump idler gear install was moved to verify thread locker is **not** disturbed for 15 minutes after initial run down (to prevent compromising thread-locker).
- Torque Value was increased for fuel pump idler gear from 30 n•m [22 ft-lb]+60° to 35 n•m [26 ft-lb]+60°

Reason for Change

In 2017, a solid lower idler gear, and adjustable fuel pump idler gear shaft were introduced. Several changes were put in place to accomodate this change:

- Factory began using gears with tape as an assembly aid. This eliminated need for feeler gages to set lash.
- New bolts with thread-locker are now specified for fuel pump idler gear when reinstalling gear to increase bolt retention margin.

- Fuel pump idler gear installation steps were moved to highlight the importance of the thread locker not being disturbed for 15 minutes after initial run down.
- Torque value for fuel pump idler gear bolts was increased from 30 n•m [22 ft-lb]+60° to 35 n•m [26 ft-lb]+60° to align with factory practices.

Service Instructions

See Table 2 for engine specific service procedure.

Service Parts Availability

Thread-locker is pre-applied to the capscrew.

Service parts are available. See Table 1 for part numbers.

Table 1, Service Parts				
Part Description	Existing Part Number	Obsolete	Superseded	New Part Number
Screw, Twelve Point Cap	3882922	Yes	Yes	4374074

Publications Affected

Table 2, Publications Affected					
Manual Type	Engine	Bulletin Number	Procedure Title	Procedure	Section
Service Manual	QX15 CM2250 ECF	2883557	Engine Base Timing	Refer to Procedure 001-088 (/qs3/pubsyss2/xml/en/procedures/132/132-001-088.html)	AS
Service Manual	ISX15 CM2250	4022250	Engine Base Timing	Refer to Procedure 001-088 (/qs3/pubsyss2/xml/en/procedures/132/132-001-088.html)	
Service Manual	Power Gen QX15 CM2250 ECF	4310661	Engine Base Timing	Refer to Procedure 001-088 (/qs3/pubsyss2/xml/en/procedures/132/132-001-088.html)	
Service Manual	Power Gen QX15 CM2250	4310664	Engine Base Timing	Refer to Procedure 001-088 (/qs3/pubsyss2/xml/en/procedures/132/132-001-088.html)	
Service Manual	ISX15 CM2350 X101	4310641	Engine Base Timing	Refer to Procedure 001-088 (/qs3/pubsyss2/xml/en/procedures/173/173-001-088.html)	

Table 2, Publications Affected					
Manual Type	Engine	Bulletin Number	Procedure Title	Procedure	Section
Service Manual	ISX15 CM2250 SN	4310736	Engine Base Timing	Refer to Procedure 001-088 (/qs3/pubsyss2/xml/en/procedures/175/175-001-088.html)	
Service Manual	QSX15 CM2350 X106	4332712	Engine Base Timing	Refer to Procedure 001-088 (/qs3/pubsyss2/xml/en/procedures/221/221-001-088.html)	
Service Manual	QSX15 CM2350 X105	4332667	Engine Base Timing	Refer to Procedure 001-088 (/qs3/pubsyss2/xml/en/procedures/250/250-001-088.html)	
Service Manual	X15 CM2350 X114 B - Efficiency Series	5411181	Engine Base Timing	Refer to Procedure 001-088 (/qs3/pubsyss2/xml/en/procedures/377/377-001-088.html)	
Service Manual	QSX15 CM2250 X115	4388739	Engine Base Timing	Refer to Procedure 001-088 (/qs3/pubsyss2/xml/en/procedures/431/431-001-088.html)	

Table 2, Publications Affected					
Manual Type	Engine	Bulletin Number	Procedure Title	Procedure	Section
Service Manual	X15 CM2350 X116 B - Performance Series	5411186	Engine Base Timing	Refer to Procedure 001-088 (/qs3/pubsyss2/xml/en/procedures/453/453-001-088.html)	
Service Manual	QSX15 CM2350 X118	5467247	Engine Base Timing	Refer to Procedure 001-088 (/qs3/pubsyss2/xml/en/procedures/485/485-001-088.html)	
Service Manual	QSX15 CM2250 ECF	2883557	Gear Train Backlash, Front	Refer to Procedure 001-055 (/qs3/pubsyss2/xml/en/procedures/132/132-001-055.html)	
Service Manual	ISX15 CM2250	4022250	Gear Train Backlash, Front	Refer to Procedure 001-055 (/qs3/pubsyss2/xml/en/procedures/132/132-001-055.html)	
Service Manual	Power Gen QSX15 CM2250 ECF	4310661	Gear Train Backlash, Front	Refer to Procedure 001-055 (/qs3/pubsyss2/xml/en/procedures/132/132-001-055.html)	

Table 2, Publications Affected					
Manual Type	Engine	Bulletin Number	Procedure Title	Procedure	Section
Service Manual	Power Gen QSX15 CM2250	4310664	Gear Train Backlash, Front	Refer to Procedure 001-055 (/qs3/pubsyss2/xml/en/procedures/132/132-001-055.html)	
Service Manual	ISX15 CM2350 X101	4310641	Gear Train Backlash, Front	Refer to Procedure 001-055 (/qs3/pubsyss2/xml/en/procedures/173/173-001-055.html)	
Service Manual	ISX15 CM2250 SN	4310736	Gear Train Backlash, Front	Refer to Procedure 001-055 (/qs3/pubsyss2/xml/en/procedures/175/175-001-055.html)	
Service Manual	QSX15 CM2350 X106	4332712	Gear Train Backlash, Front	Refer to Procedure 001-055 (/qs3/pubsyss2/xml/en/procedures/221/221-001-055.html)	
Service Manual	QSX15 CM2350 X105	4332667	Gear Train Backlash, Front	Refer to Procedure 001-055 (/qs3/pubsyss2/xml/en/procedures/250/250-001-055.html)	

Table 2, Publications Affected					
Manual Type	Engine	Bulletin Number	Procedure Title	Procedure	Section
Service Manual	X15 CM2350 X114 B - Efficiency Series	5411181	Gear Train Backlash, Front	Refer to Procedure 001-055 (/qs3/pubsys2/xml/en/procedures/377/377-001-055.html)	
Service Manual	QSX15 CM2250 X115	4388739	Gear Train Backlash, Front	Refer to Procedure 001-055 (/qs3/pubsys2/xml/en/procedures/431/431-001-055.html)	
Service Manual	X15 CM2350 X116 B - Performance Series	5411186	Gear Train Backlash, Front	Refer to Procedure 001-055 (/qs3/pubsys2/xml/en/procedures/453/453-001-055.html)	
Service Manual	QSX15 CM2350 X118	5467247	Gear Train Backlash, Front	Refer to Procedure 001-055 (/qs3/pubsys2/xml/en/procedures/485/485-001-055.html)	

Document History

Date	Details
2018-4-25	Module Created

